tf.clip_by_global_norm Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/clip_by_global_norm

> *Clips values of multiple tensors by the ratio of the sum of their norms.*

Function Signatures

```
tf.clip_by_global_norm( t_list, clip_norm, use_norm=None,
name=None )
```

Code Examples

```
tf.clip_by_global_norm(
t_list, clip_norm, use_norm=None, name=None
)
```


```
tf.clip_by_global_norm(
t_list, clip_norm, use_norm=None, name=None
)
```


```
t_list[i] * clip_norm / max(global_norm, clip_norm)
```


```
t_list[i] * clip_norm / max(global_norm, clip_norm)
```


```
global_norm = sqrt(sum([l2norm(t)**2 for t in t_list]))
```


```
global_norm = sqrt(sum([l2norm(t)**2 for t in t_list]))
```


Documentation

Clips values of multiple tensors by the ratio of the sum of their norms.

View aliases

Compat aliases for migration

See
Migration guide for
more details.

tf.compat.v1.clip_by_global_norm

Given a tuple or list of tensors t_list, and a clipping ratio clip_norm,
this operation returns a list of clipped tensors list_clipped
and the global norm (global_norm) of all tensors in t_list. Optionally,
if you've already computed the global norm for t_list, you can specify
the global norm with use_norm.

To perform the clipping, the values t_list[i] are set to:

where:

If clip_norm > global_norm then the entries in t_list remain as they are,
otherwise they're all shrunk by the global ratio.

If global_norm == infinity then the entries in t_list are all set to NaN
to signal that an error occurred.

Any of the entries of t_list that are of type None are ignored.

This is the correct way to perform gradient clipping (Pascanu et al., 2012).

However, it is slower than clip_by_norm() because all the parameters must be
ready before the clipping operation can be performed.

Returns

Raises

References